

LESSON 15 / 第十五課

Veteran Trees: From Ancient Populations to Cambial Islands

老齡樹：從古老族群到形成層島

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

A heritage or veteran register is a status list, not a diagnosis. Veteran physiology is retrenchment and reiteration: Lonsdale, Read and Fay described functional units; Ishii documented reiterated shoots in old crowns. Living wood may already be islands. Heavy live-dose pruning spends the NSC of those islands. Dual consequence applies: the tree as habitat and culture, and the tree as a target-bearing structure. Staged dose protects remaining cambial strips. Tropical experimental carbon budgets are thinner than European practice texts.

遺產或老齡名錄是身份名單，不是診斷。老齡生理是回縮與再迭代：Lonsdale、Read 與 Fay 描述功能單元；Ishii 記錄老齡樹冠中的再迭代枝。活木材可能已成島。重活劑量修剪花掉那些島的非結構性碳水化合物。雙重後果適用：樹作為棲地與文化，以及樹作為承載目標的結構。分期劑量保護剩餘形成層條帶。熱帶實驗碳預算薄於歐洲實務文本。

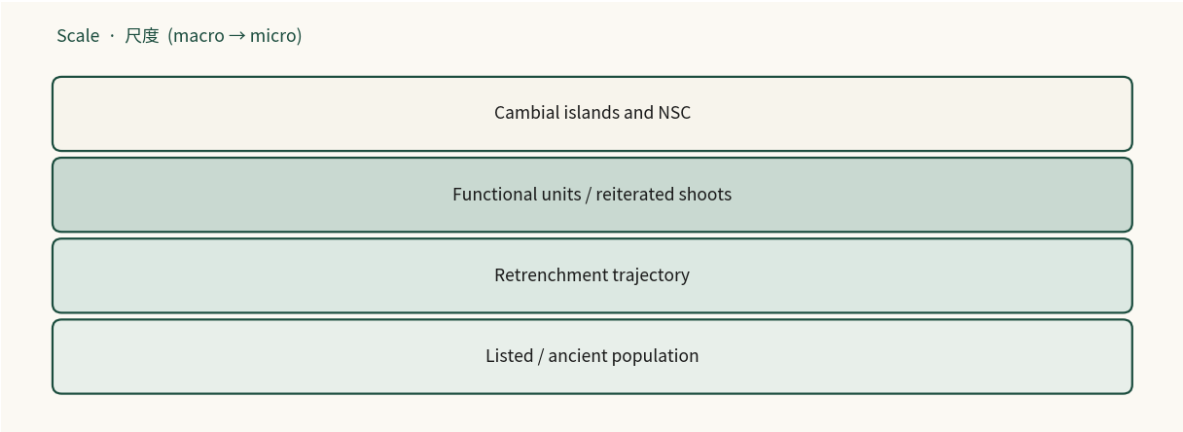


Figure 1. Nested scales from listed populations to cambial islands.

圖 1. 由登錄族群到形成層島的嵌套尺度。

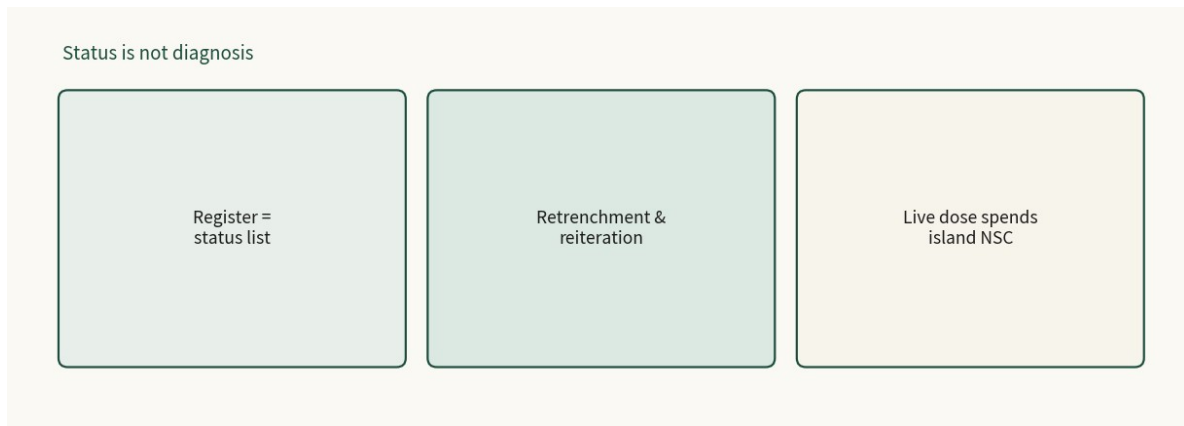


Figure 2. Register, trajectory and live dose are different objects.

圖2. 名錄、軌跡與活劑量是不同對象。

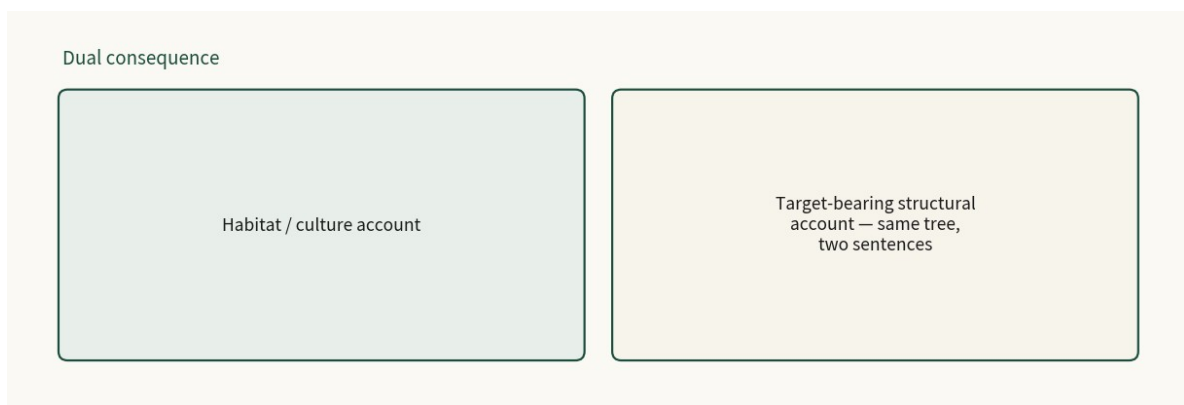


Figure 3. Dual consequence: habitat and structure.

圖3. 雙重後果：棲地與結構。

1. Scope

1. 範圍

The object is published veteran-tree biology, from listed populations to remaining cambium. A local register's five criteria are an example of status, not a decay map.

對象是已發表老齡生物學：由登錄族群到剩餘形成層。地地名錄的五項準則是身份的例子，不是腐朽圖。

2. Trajectory and units

2. 軌跡與單元

Retrenchment and reiterated shoots are ageing, not failure per se [1–3]. Lonsdale's functional units and Ishii's reiteration describe semi-autonomous living parts that can persist after the original crown has shortened [1,3]. Listing mixes size, age, culture and habitat; it is not a risk grade [4].

回縮與再迭代枝是老化，不是失效本身 [1–3]。Lonsdale 的功能單元與 Ishii 的再迭代，描述原樹冠縮短之後仍可存續的半自主活體部分 [1,3]。登錄混合尺寸、年齡、文化與生境；不是風險等級 [4]。

3. Dose and dual consequence

3. 劑量與雙重後果

Heavy live pruning spends NSC from remaining strips. Staged dose is the published precaution where the carbon budget is unknown [1,2]. The same tree carries two consequence accounts: habitat and culture, and a target-bearing structure. Neither account abolishes the other.

重活修剪花掉剩餘條帶的 NSC。在碳預算未知之處，分期劑量是已發表的審慎做法 [1,2]。同一棵樹承載兩本後果賬：棲地與文化，以及承載目標的結構。任一賬都不取消另一賬。

4. Limits

4. 限度

European practice texts are thicker than tropical experimental carbon budgets. Copying a veteran-oak dose onto a tropical fig island is a transfer, not a measurement.

歐洲實務文本厚於熱帶實驗碳預算。把老齡櫟劑量抄到熱帶榕島上，是轉移而不是測量。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. A register is:

1. 名錄是：

- A. A diagnosis.
- A. 診斷。
- B. A status list.
- B. 身份名單。
- C. P50.
- C. P50。
- D. A tomogram verdict.
- D. 斷層判決。

2. Retrenchment is:

2. 回縮是：

- A. Instant failure.
- A. 立即失效。
- B. Part of ageing.
- B. 老化的一部分。
- C. A soil test.
- C. 土壤檢測。
- D. 3-30-300.
- D. 3-30-300。

3. Functional units are:

3. 功能單元是：

- A. Policies.
- A. 保單。
- B. Semi-autonomous living parts.

B. 半自主活體部分。

C. Cable sizes.

C. 鋼纜尺寸。

D. Flash maps.

D. 閃擊圖。

4. Heavy live pruning:

4. 重活修剪：

A. Is free.

A. 免費。

B. Spends island NSC.

B. 花掉島的 NSC。

C. Restores youth.

C. 恢復青春。

D. Sets P12.

D. 設定 P12。

5. Listing mixes:

5. 登錄混合：

A. Paint only.

A. 只有漆。

B. Size, age, culture, habitat.

B. 尺寸、年齡、文化、生境。

C. Only t/R.

C. 只有 t/R。

D. Only i-Tree.

D. 只有 i-Tree。

6. Reiteration:

6. 再迭代：

A. Never occurs.

A. 從不發生。

B. Is documented in old crowns.

B. 在老齡樹冠有記錄。

C. Is lightning.

C. 是雷電。

D. Equals PLC.

D. 等於 PLC。

7. Tropical veteran C budgets:

7. 熱帶老齡碳預算：

A. Thicker than EU practice.

A. 厚於歐實務。

B. Thinner.

B. 較薄。

C. Complete.

C. 完整。

D. Illegal.

D. 非法。

8. Dose should be:

8. 劑量應當：

A. Maximal.

A. 最大。

B. Staged.

B. 分期。

C. Random.

C. 隨機。

D. 10-20-30.

D. 10-20-30。

9. Habitat and structural accounts:

9. 棲地賬與結構賬：

A. Abolish each other.

A. 互相取消。

B. Both remain in view.

B. 兩者都留在視野內。

C. Equal P88.

C. 等於 P88。

D. Are mulch rules.

D. 是覆蓋規則。

10. Copying a veteran-oak dose onto a tropical fig is:

10. 把老齡櫟劑量抄到熱帶榕上是：

A. A measurement.

A. 一項測量。

B. A transfer.

B. 一項轉移。

C. Wall 2.

C. 第 2 壁。

D. A flare datum.

D. 幹基基準。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Lonsdale D. veteran physiology and functional units.

[2] Read H., Fay N. veteran-tree handbooks.

[3] Ishii H. reiteration in old crowns.

[4] Comparative heritage / veteran listing criteria.

[5] Fay N. veteran-tree management and staged reduction.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。